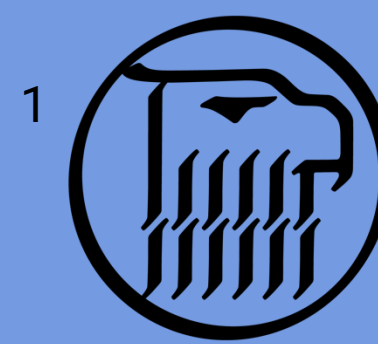


GC53C–0400: Risk analysis for the energy sector via the Cal–Adapt Analytics Engine: A co–production approach for tailored climate analytics in a warming world

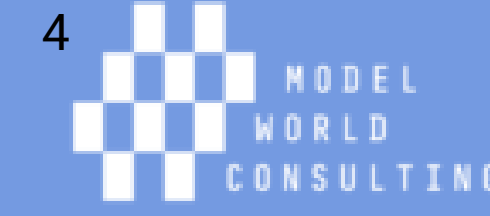
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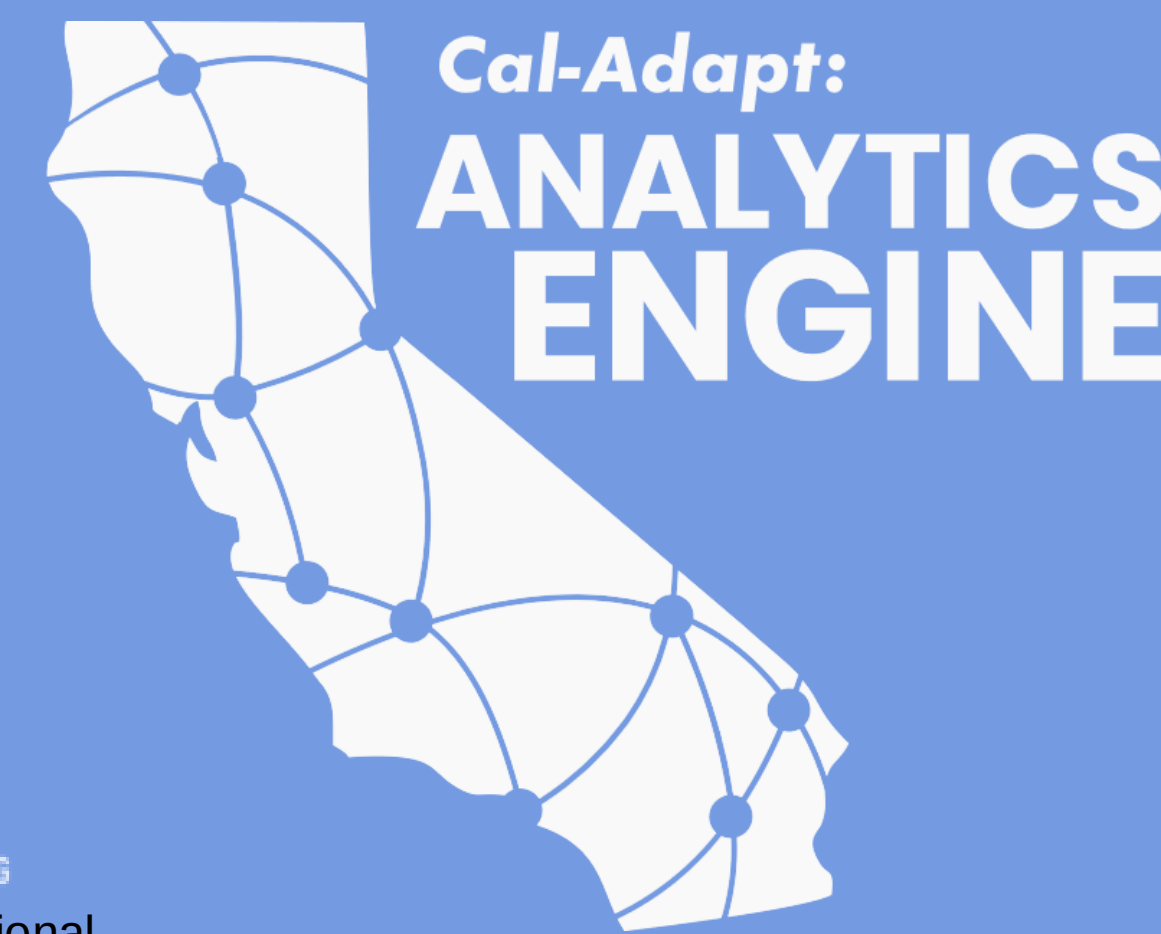
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Data for Decisions

Vulnerability assessments are a systematic review to evaluate the exposure of resources and people to climate impacts, and the resulting vulnerability to climate risks and hazards. Vulnerability assessments:

- **Identify vulnerabilities** and the level of risk involved
- Provide **recommendations** for how to address vulnerabilities
- Can be **highly time intensive** based on the level of specificity required
- Can be **required** or **legally mandated** for organizations, state agencies

Energy Partner Needs

California Investor Owned Utilities (IOUs) are legally mandated by the California Public Utility Commission (CPUC) to conduct vulnerability assessments. Challenges include:

1. In 2024, the CPUC adopted **requirements that IOUs must use global warming levels** for climate vulnerability analyses, replacing a time-based planning horizon approach. *The Analytics Engine was identified as the **recommended source** for vetted climate data and guidance¹.*
2. **Clear, comprehensive climate data guidance** and guidelines for how climate data should and should not be used is a high priority for the next generation of climate-informed vulnerability assessments².

Assessing risk within a warming world is hard, doing so within a regulatory environment is even harder.

Co-Producing Climate-Informed Processes with Energy Partners

The Cal-Adapt: Analytics Engine provides a flexible environment for analyzing and processing climate data on the cloud, making data available to decision makers in the form desired for their application. Collaboration via co-production develops trusted relationships through facilitated conversations between scientists, developers, and energy-sector practitioners, bridging climate expertise and grid resilience experts.

Initial Pilot

Exposure analysis tools may support IOU vulnerability assessments through asset planning and strategy, and proposed project designs to account for future climates. Through co-production, we developed a pipeline to calculate fully customizable exposure metrics, centered on the tenant that “a warming level based approach does not require users to fundamentally change downstream vulnerability analysis”.

A Vulnerability Assessment Pipeline for Data-Informed Solutions

Example decision to build custom data metrics for exposure analysis:

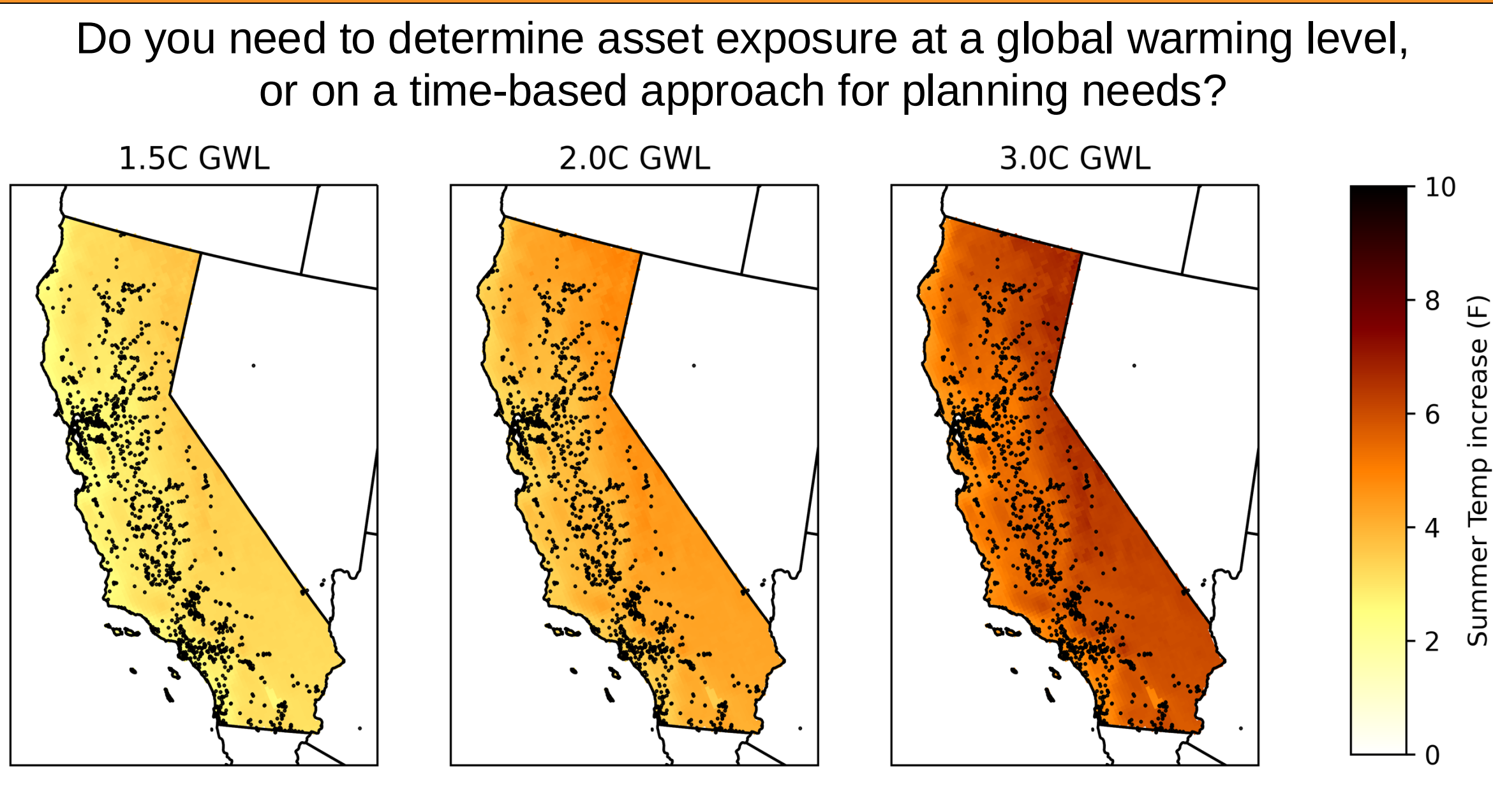


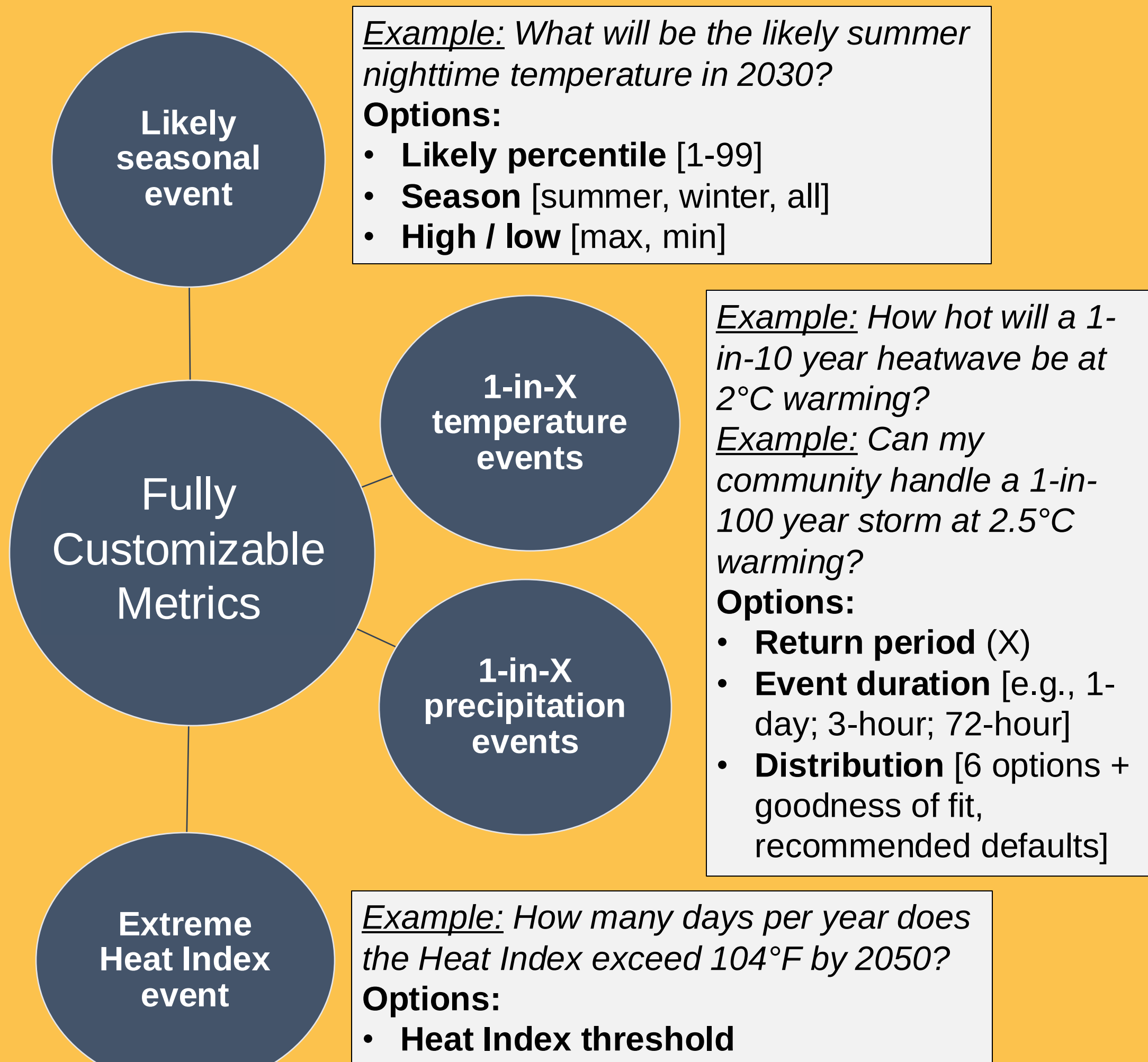
Figure 1. Multi-model mean average summer air temperature increase across California, in °F at global warming levels (1.5°C, 2.0°C, and 3.0°C) in the dynamically downscaled 3-km WRF models for CA. California Energy Commission (CEC) power-plants and substations are overlain as an example of asset distribution in a vulnerability assessment exposure analysis. Produced on the Cal-Adapt: Analytics Engine.

Transparent + Flexible Solutioning

Co-production for the Vulnerability Assessment framework included:

- Biweekly 1-on-1 check-ins with utility partner in active development of their upcoming mandatory vulnerability assessment
- Iterative feedback to ensure development consistently meets partner needs while building in flexibility for future use cases
- Working group demo to state and federal stakeholders
- Robust testing + technical support, through usability testing
- Methodological support + documentation of open-source Jupyter notebook³ and code⁴

V1 of the fully functional exposure analysis tool can lead to data-driven solutions for the following types of questions:



Additional options for all metrics

of locations (single or many), optimized for multiple locations

- Recommendations for size limits provided + approx. runtimes

Approach

- **Warming levels:** 0.8, 1.2, 1.5, 2.0, 2.5, 3.0 °C options
- **Time-based:** Fully flexible start and end years, SSPs [LOCA2]
- **Downscaled GCM data:** Dynamical [WRF, WRF-ERA5], Statistical [LOCA2] data produced in support of CA's 5th Climate Change Assessment ^{5,6}

- Option to return only bias-adjusted WRF models

Units: [Temp: degC, degF, K. Precip: mm, inches]

Export method: Raw data input, computed metric, both

File format: NetCDF, csv + metadata

Practical Applications

Uses of climate data are highly varied. Through the iterative development process, **11 realistic scenarios** of data processing were provided in the vulnerability assessment Jupyter notebook to directly support the broad kinds of questions facing utility partners in plain language, including:

- “I want to calculate the likely summer day high temperature with a 2°C warming level, where likely is the 50th percentile, in degC, using all available WRF data” (**Figure 2**)
- “I want to calculate the 1-in-10 year heatwave event using a GEV distribution, in degC, for a 2.5°C warming level with LOCA2 data”
- “I want to calculate the 1-in-100 year precipitation event for a 3-hour event using the GEV distribution, in inches, for 2070-2090, with SSP3-7.0 using only the bias-adjusted WRF data”
- “I want to calculate the number of days per year that the heat index exceeds 104°F between 2030-2060”

Example: Global Warming Level approach with WRF

Example scenario: I want to calculate “likely summer day high temperature with a 2°C warming level, where likely is the 50th percentile, in Celsius, using all available WRF data (bias-adjusted and non-bias-adjusted), and export both the raw and calculated metric data”. I would input:

```
data = cava_data(  
    ## Set-up  
    example_locs.iloc[:1], # select a single location  
    downscaling_method="Dynamical", # WRF data  
    approach="Warming Level",  
    warming_level=2.0,  
    wrf_bias_adjust=False, # return all WRF models  
  
    ## Likely seasonal event specific arguments  
    variable="Air Temperature at 2m",  
    metric_calc="max", # daily high temperature  
    percentile=50, # likelihood percentile  
    season="summer", # season  
    units="degC", # change units  
  
    ## Export  
    export_method="both",  
    file_format="NetCDF",  
)
```

Last executed at 2024-12-06 15:13:33 in 1m 21.14s

Calculating 50th percentile of Daily Max of Air Temperature at 2m for summer seasons for Warming Level 2.0°C

--- Selecting Data Points ---

Selecting data for (35.43424, -119.05524)

--- Loading Data into Memory ---

Point (35.445845471191406, -119.06060791015625)

Processing data to read 2.02 MB of data into memory...

[#####] | 100% Completed | 64.42 s Complete!

Figure 2. Example usage of the `cava_data` function⁴ which underpins the vulnerability assessment Jupyter notebook. Example plain-language scenario is provided with user-friendly options explained in the function call. Providing clearly documented examples of realistic problems facing utility partners helps build confidence in the tool and notebook, as well as building capacity for climate data access.

Climate Science Approach

A core proponent of the co-production process was to **provide a dedicated, welcoming space** for partners to ask questions relevant to their workflows, including the **scientific best practices** and **understanding of the complexity of the climate data space** (e.g., the calculation and application of *global warming levels*).

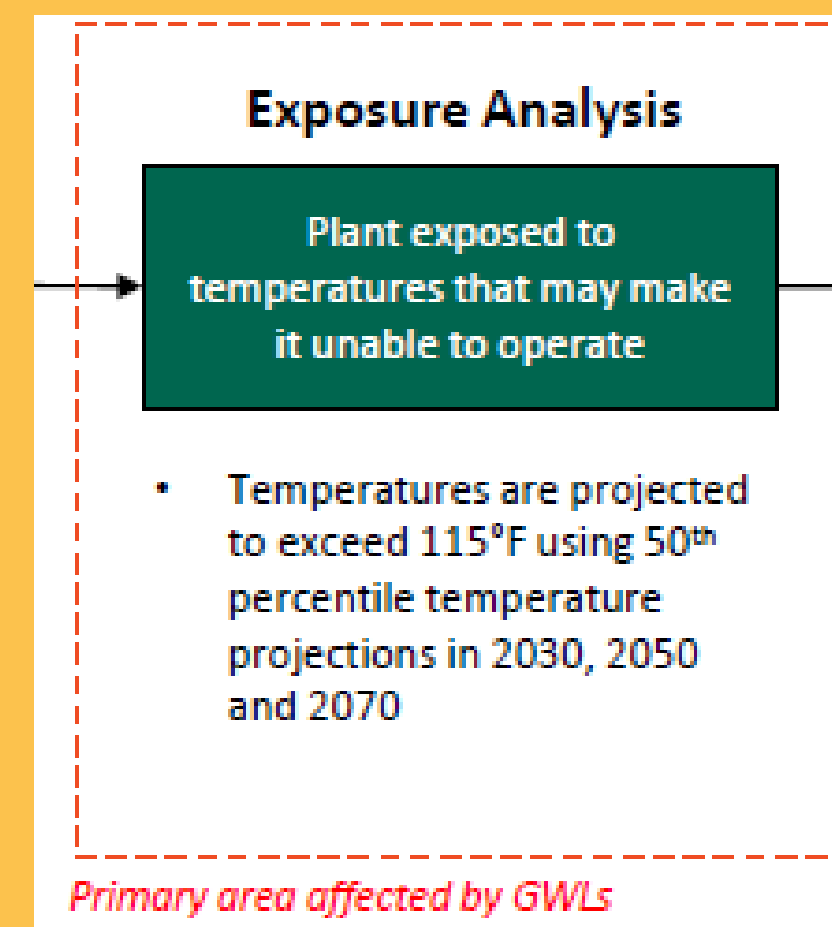


Figure 3. Diagram of hypothetical process incorporating warming levels.

In response to the need for clarity and guidance, we provide **technical methods documentation**, example usage, arguments and their definitions, order of operations, and any relevant processing notes with statistical guidance:

- Data limitations (WRF vs. LOCA2)
- Goodness of fit and recommended first pass statistical distribution for 1-in-X event analysis
- Limitations of data sample size in realistic geographic regions (e.g., 1-in-1000 year storm in a desert)

Iteration and Feedback is Key

Engagement is ongoing – feedback, regular communication, and improved iteration is critical for the successful application of a vulnerability assessment.

Building user capacity and technological / scientific innovation within a regulatory environment is essential to overcome barriers in the vulnerability assessment space.

What's coming next?

Enhanced multiple location support

- **Goal:** A user should not be limited by the number of locations

Incorporating uncertainty and confidence ranges

- **Goal:** Building further capacity for users to understand the complexity of climate data and form robust, confident decisions

Different data options

- **Goal:** Incorporating station-based data, hydrology data

Additional metrics

- **Iteration through engagement and co-production:**
 - What are key vulnerability assessment related questions that you are working with?
 - What metrics or thresholds are important for your vulnerability assessments?
 - What outputs or products are of the most interest to you?

Dedicated efforts to develop guidance alongside state agencies and build credibility with end users will help encourage greater adoption of these data and tools.

Contact us!

analytics@cal-adapt.org

Scan the QR code below to visit us at
analytics.cal-adapt.org



Thank you to our funders



California Energy Commission – EPC-20-007 & EPC-23-024

References and Cal-Adapt at AGU

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3. https://github.com/cal-adapt/cae-notebooks/collaborative/IOU/vulnerability_assessment
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5. Rahimi et al. 2024. <https://doi.org/10.1029/2023GL106264>
6. Pierce et al. 2023. <https://doi.org/10.1175/JHM-D-22-0194.1>

Interested in more of the Cal-Adapt work? Check out these presentations:

GC21E: Co-producing guidance on the use of climate data for scientific and decision applications [Tues 830am]
GC23E: Navigating practice, science, and regulation – a regional framework for climate services based on lessons from the Cal-Adapt: Analytics Engine [Tues 1:40pm]
TH33H: Climate Information and Services: A Whole-of-Government Approach [Weds 1230pm]